
LC Electromagnetism I Lecture Notes

Year 1 Semester 2

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Lecture 1 - Electric Fields I: EM1 Introduction

1 Course Intro

Course Materials:

- Background material and derivations etc on PowerPoint.
- Worked examples etc are handwritten on visualiser, these are the bits we really need to know.

Why is EM important?

- Foundations of modern technology and the modern world.
- What gives elements their properties.
- Responsible for life itself.
- Everyday materials are held together by EM forces.
- Optics can only be understood through EM theory.

The course aim is to lay down the foundations, eventually leading us to Maxell's Laws.

1.1 Maxwell's Laws

Maxwell's four equations are:

$$\begin{aligned}\nabla \cdot \mathbf{E} &= \frac{\rho}{\epsilon_0} \\ \nabla \wedge \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t} \\ \nabla \cdot \mathbf{B} &= 0 \\ \nabla \wedge \mathbf{B} &= \mu_0 \left(\mathbf{J} + \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right)\end{aligned}$$

Where:

$$\nabla = \mathbf{i} \frac{\partial}{\partial x} + \mathbf{j} \frac{\partial}{\partial y} + \mathbf{k} \frac{\partial}{\partial z}$$

Together, these show that the electric and magnetic fields are related and are two aspects of a single force, the electromagnetic force. We don't have to properly understand them yet, but cannot learn them in EMII unless we fundamentally understand E and B fields from this module.

1.2 Course Structure

Part I: Electric Fields

- Charge and Coulomb's Law.
- The electric field.
- Gauss' Law.
- Capacitors.

Part II: Magnetic Fields

- Magnetic Fields

- Charged Particles in B-Fields
- Electromagnetic Induction.
- Magnetic Dipoles

In this lecture:

- Introduction to EM.
- Electric charge.
- Force between charges.
- The concept of the Electric Field (E-Field).

2 Electric Charge

First attributed to Thales circa. 624 - 546 BC. Experiments by Franklin and Coulomb expanded and showed that there was two types of charge, which they called positive and negative.

The “positive electricity” came from rubbing a glass rod with silk, and the negative from rubbing an ebonite (early plastic) rod with fur. They found that like charges repel and opposite charges attract.

We know that the elementary unit charge is the magnitude of charge of an electron/proton and everything else is a multiple of this ¹:

$$e = 1.6 \times 10^{-19} C$$

This has units of the Coulomb.

2.1 Charge Conservation

Electrons and protons are both stable (protons decay with a life greater than 10^{31} years). This means that the total charge of an isolated system is constant and can be conserved.

They have the same magnitude of charge, exactly:

$$|q_p| = |q_e| = e$$

2.2 Electrostatic Force

Like charges repel and opposite charges attract, along the line of action given by a line drawn between the two charges. The force is proportional to the product of charges so:

$$F \propto q_1 q_2$$

Here, a negative force means attraction and a positive force means repulsion. Newton called this “force at a distance”. Like gravity, two charges will exert a force on each other at a distance without any contact.

There must, therefore, be something between them that mediates this force. Later physics gives this as “virtual particles” which isn’t a Y1 topic, so classically we say that this medium is the Electric Field.

2.3 Electric Field

A charge produces a field around it. Another charge also interacts with this field, and this interaction is what causes a force:

$$\underline{F} = \underline{E}q$$

Where F is the force exerted on a test charge of charge q by a charge Q producing a field E . The magnitude of the electric field has units NC^{-1} (force per units charge).

$$|\underline{E}| \propto Q \quad |\underline{F}| \propto Qq$$

¹While quarks have fractional charge, we don’t get free quarks

Consider a point charge with a spherical electric field spreading out around it. As the distance from the charge increases, the surface area increases as $4\pi r^2$. Therefore the magnitude of the electric field must decrease with $4\pi r^2$

Therefore:

$$|\underline{E}| \propto \frac{Q}{4\pi r^2}$$

We need a (inverse) constant of proportionality. This depends on the medium, but for a vacuum we call it the permittivity of free space ϵ_0 :

$$\epsilon_0 = 8.854 \times 10^{-12} C^2 m^{-2} N^{-1}$$

Hence:

$$E = |\underline{E}| = \frac{Q}{4\pi\epsilon_0 r^2}$$

2.4 Direction of the E-Field

Force is a vector, so the E-field must be too. Consider an E-field from a charge Q at distance r . We give E components E_r , E_θ , E_ϕ , where, since it's a sphere ϕ and θ represent the unit vectors in the two possible tangential directions.

If there was a component in E_θ this would be clockwise from one perspective, but anticlockwise from another (walking behind it). This is not possible, as the field must behave in the same manner from all viewpoints. Therefore:

$$E_\theta = E_\phi = 0$$

$$\underline{E} = \frac{Q}{4\pi\epsilon_0 r^2} \hat{r}$$

2.5 Force between two charges

Consider two charges q_1 and q_2 . The force on q_2 due to the e-field from q_1 is:

$$\underline{F}_1 = \underline{E}_1 q_2$$

This is equal to the force on q_1 due to the e-field from q_2 , given by:

$$\underline{F}_2 = \underline{E}_2 q_1$$

So the force between two charges is:

$$\underline{F} = \frac{q_1 q_2}{4\pi\epsilon_0 r^2} \hat{r}_{12}$$

2.6 Force between many charges

If we have more than two positive charges, we use the “principle of superposition”. Effectively, you consider each pair of charges at the same time and vector sum of the forces together. I.e. if we have three points and we care about the net force on one, we take the vector sum of the two vectors from that point to the others.

In general:

$$\begin{aligned} \underline{F} &= \underline{F}_1 + \underline{F}_2 + \underline{F}_3 + \dots \\ &= q \sum_i \frac{q_i}{4\pi\epsilon_0 r_i^2} \hat{r}_i \end{aligned}$$

Where r_j is the distance between q_i and q , with unit vector $\hat{r}_j$ between them.

Since $\underline{F} = q\underline{E}$, the electric field at a test charge q must be:

$$\underline{E} = \sum_i \frac{q_i}{4\pi\epsilon_0 r_i^2} \hat{r}_i$$

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Lecture 2 - Electric Fields II: Field Lines and Continuous Charges

1 Field Lines

To visualise a field, we draw field lines. These are drawn so that the tangent of these lines at any point in space gives the direction of $\underline{E}$ at that point.

The lines point radially outwards from a positive charge, and radially towards a negative charge. The density of these lines is proportional to the magnitude of $\underline{E}$, but the absolute number of lines doesn't convey much.

Field lines must start at a positive charge and end on a negative charge. They are continuous and can never cross each other, they only start/stop at charges or proceed to infinity.

2 Continuous Charge Distributions

What do we do if we have a charge which isn't a point particle?

- We break the object up into a distribution of infinitesimal point charges.
- We treat each piece as a point charge and add up the separate $\underline{E}$ fields across an integral.
- This gets messy for anything non-trivial, resulting in a complex 3D integral often requiring computational techniques.

Charges can be distributed across an object in a number of ways:

- Charge per unit length across a line, given by λ :

$$q = \int \lambda dl$$

- Charge per unit area across a surface area, given by σ :

$$q = \int \sigma dA$$

- Charge per unit volume, given by ρ :

$$q = \int \rho dV$$

3 Example: E-Field from a Line of Charge

Consider an E-field from a line of length $2L$, measured at a point along the midpoint's perpendicular. This point sits x length units along the perpendicular.

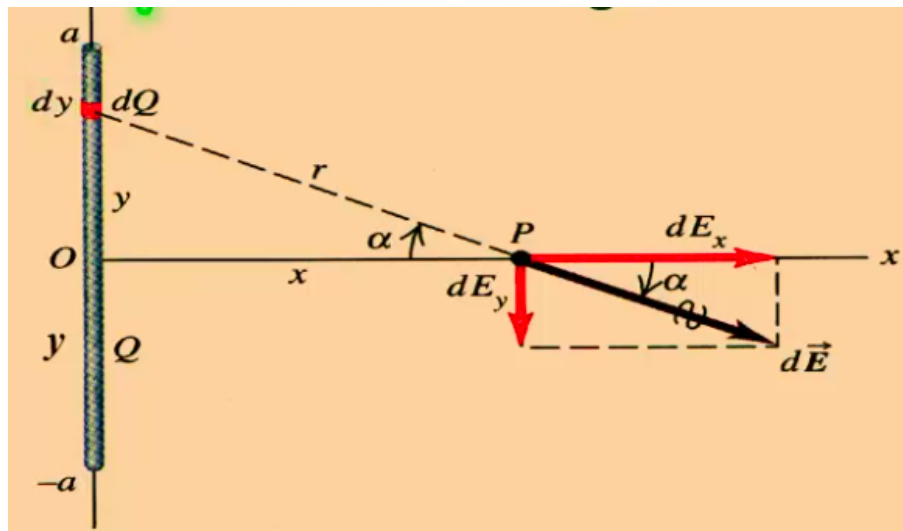


Figure 2.1

We consider some element of charge $\delta Q = \lambda \delta y$ along the line at y . There are two components to the E-field, one pointing in the positive x direction and one in the negative y direction.

When we consider all of the δQ s along the length of the charged rod, each point on the positive y -axis along the length will have a corresponding point on the negative y -axis. These two points will have equal and opposite magnitude, so $E_y = 0$ by symmetry. There is therefore no resultant y component and the final field will point along $\hat{x}$.

For some infinitesimal δq generically along the line of charge:

$$\delta E_x = \frac{\delta q}{4\pi\epsilon_0 r^2} \cos \alpha \quad (1)$$

And for each δQ :

$$\delta Q = \lambda \delta y \quad (2)$$

And from trig:

$$r^2 = y^2 + x^2 \quad (3)$$

$$\cos \alpha = \frac{x}{r} \quad (4)$$

Substituting 2, 3, 4 into 1:

$$\delta E_x = \frac{\lambda \delta y}{4\pi\epsilon_0} \frac{x}{[y^2 + x^2]^{3/2}}$$

Hence:

$$E_x = \frac{\lambda x}{4\pi\epsilon_0} \int_{-L}^L \frac{1}{[y^2 + x^2]^{3/2}} dy$$

To solve this, we would use the standard integral given in the question. Alternatively, if a standard integral isn't given in the question we could express in the integral in terms of α and integrate from the minimum to maximum α .

3.1 Example Part B: The Easier Route

As before:

$$\delta E_x = \frac{\lambda \delta y}{4\pi\epsilon_0 r^2} \cos \alpha \quad (1)$$

$$\cos \alpha = \frac{x}{r} \implies r = \frac{x}{\cos \alpha} \quad (2)$$

And:

$$\tan \alpha = \frac{y}{x} \implies y = x \tan \alpha \quad (3)$$

As:

$$\frac{d}{d\alpha} \tan \alpha = \frac{1}{\cos^2 \alpha}$$

We have:

$$\delta y = \frac{x \delta \alpha}{\cos^2 \alpha} \quad (4)$$

Hence:

$$\begin{aligned} \delta E_x &= \frac{\lambda}{4\pi\epsilon_0} \underbrace{\frac{x \delta x}{\cos^2 \alpha}}_{\delta y} \underbrace{\frac{\cos^2 \alpha}{x^2}}_{1/r^2} \cos \alpha \\ &= \frac{\lambda}{4\pi\epsilon_0 x} \cos \alpha \end{aligned}$$

Which is much easier to integrate from the minimum to maximum values of α .

$$E_x = \frac{\lambda}{4\pi\epsilon_0 x} \int_{\alpha_{\min}}^{\alpha_{\max}} \cos \alpha \, d\alpha$$

And from the geometry of the problem:

$$\alpha_{\max} = \arctan\left(\frac{L}{x}\right) \quad \alpha_{\min} = -\arctan\left(\frac{L}{x}\right)$$

Which is easy to integrate.

If the problem was setup such that the rod was infinitely long, then:

$$\begin{aligned} E_x &= \frac{\lambda}{4\pi\epsilon_0 x} \int_{-\pi/2}^{\pi/2} \cos \alpha \, d\alpha \\ E_x &= \frac{\lambda}{2\pi\epsilon_0 x} \end{aligned}$$

This is approximately true for the case where $x \ll L$.

4 Example: Ring of Charge

Consider a ring of charge with radius a . We step out some perpendicular constant distance x from the centre of the ring to point P . What is the resultant field at P ?

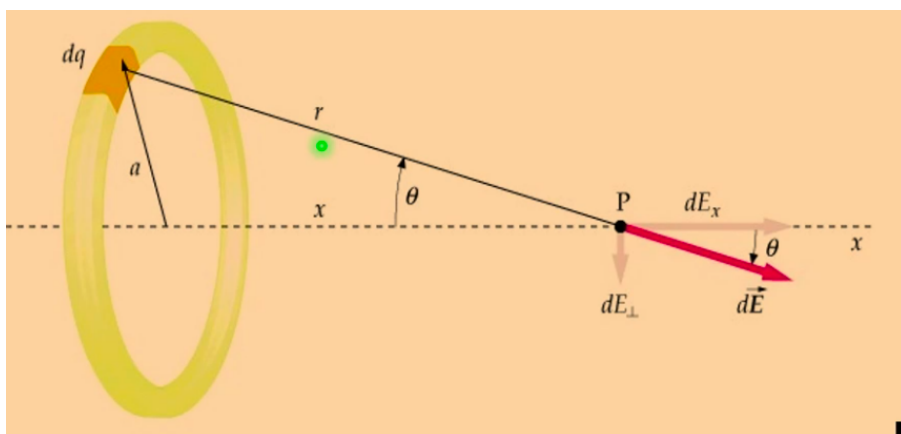


Figure 2.2

The ring has total charge Q , distributed with charge per unit length λ .

Again, we can use symmetry. For every point on the ring, there will be a point one diameter away creating an equal and opposite δE . Hence, the y and z components cancel leaving only E_x , pointing in the $\hat{x}$ direction.

$$E_y = E_z = 0 \text{ by symmetry}$$

We have:

$$\begin{aligned}\delta E_x &= \frac{\delta q}{4\pi\epsilon_0 r^2} \cos \theta \\ \cos \theta &= \frac{x}{r} \\ r &= \sqrt{a^2 + x^2}\end{aligned}$$

Hence:

$$\delta E_x = \frac{\delta q}{4\pi\epsilon_0} \frac{x}{[a^2 + x^2]^{3/2}}$$

Here, a and x are constants, so the only variable we can integrate across is δq :

$$\begin{aligned}E_x &= \frac{x}{4\pi\epsilon_0} \frac{1}{[a^2 + x^2]^{3/2}} \int dq \\ &= \frac{xQ}{4\pi\epsilon_0 [a^2 + x^2]^{3/2}}\end{aligned}$$

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Lecture 3 - Electric Fields III: More Continuous Charges and Gauss' Law

1 Example: Large Circular Plane

Consider a large circular plane with uniform charge density σ . We step out some distance x along the axis perpendicular to the plane, through its centre, to point P .

The plane has radius R and total charge Q . What is the magnitude and direction of the electric field at P ?

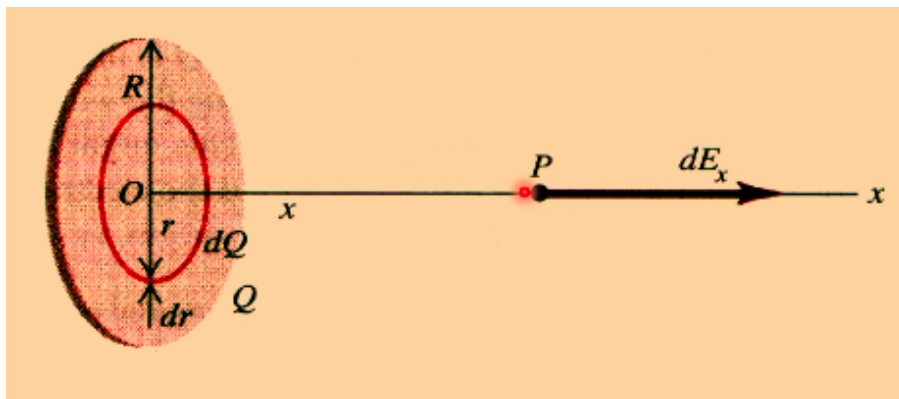


Figure 3.1

Using the same logic as for the ring last lecture, we can see that:

$$E_y = E_z = 0 \text{ by symmetry}$$

So the final E-field lies entirely in the direction of $\hat{x}$.

We want to divide the final E-field into an infinite number of thin rings, each at radius r with infinitesimal thickness δr . We can then integrate across all rings to determine the total E-field.

The distance between P and O is x , the distance between O and the charge element is r , and the distance between P and the charge element is d .

From the previous example (in an exam question we would have to do this too), we have:

$$\delta E_{\text{ring}} = \frac{Q_{\text{ring}}}{4\pi\epsilon_0} \frac{x}{[r^2 + x^2]^{(3/2)}}$$

$$Q_{\text{ring}} = \sigma \times 2\pi r \delta r$$

Hence:

$$\delta E_{\text{ring}} = \frac{\sigma r \delta r}{2\epsilon_0} \frac{x}{(r^2 + x^2)^{(3/2)}}$$

Noting that $r = x \tan \theta$, where θ is the angle between d and x :

$$\delta r = \frac{x}{\cos^2 \theta} d\theta$$

$$\Rightarrow \cos \theta = \frac{x}{d} = \frac{x}{[r^2 + x^2]^{1/2}}$$

$$\frac{1}{[r^2 + x^2]^{3/2}} = \frac{\cos^3 \theta}{x^3}$$

Substituting these in:

$$\delta E_{\text{ring}} = \frac{\sigma}{2\epsilon_0} \underbrace{\left(\frac{x \sin \theta}{\cos \theta} \right)}_r \underbrace{\frac{x \delta \theta}{\cos^2 \theta}}_{\delta r} \times x \times \frac{\cos^3 \theta}{x^3}$$

Which simplifies down to:

$$\frac{\sigma}{2\epsilon_0} \sin \theta \delta \theta$$

Which is again nice and simple to integrate. Integrating across all θ :

$$E_{\text{disc}} = \frac{\sigma}{2\epsilon_0} \int_0^{\theta_{\text{max}}} \sin \theta d\theta$$

Where $\theta_{\text{max}} = \arctan(R/x)$. If we wanted a full infinite plane, we could get this as $R \rightarrow \infty$.

$$\theta_{\text{max}} = \pi/2$$

$$E = \frac{\sigma}{2\epsilon_0} \int_0^{\pi/2} \sin \theta d\theta$$

$$\Rightarrow E = \frac{\sigma}{2\epsilon_0}$$

Similar to the ring, this is approximately true for a finite plane where $x \ll R$. This is interesting because it implies that (for close distances, where $x \ll R$ still holds), the electric field is constant with respect to distance.

2 E-Field in a Hollow Sphere

Consider a hollow sphere with uniform charged shell density σ . Consider an arbitrary point P inside this spherical shell.

We can use solid angles here, which are effectively three dimensional angles. This is still dimensionless, but is given by a subtended area divided by the radius squared:

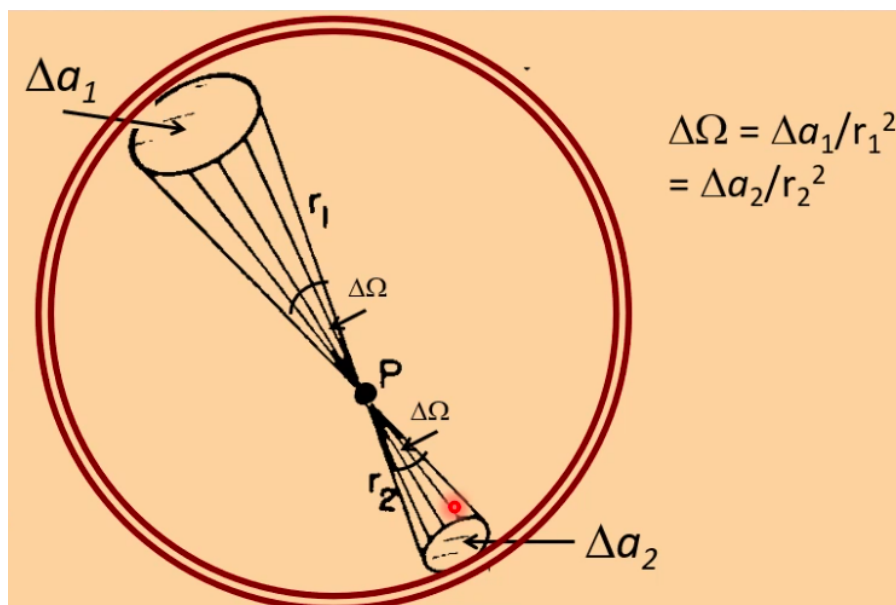


Figure 3.2

While a traditional angle is $\theta = \text{arc length} / \text{radius}$ in units of radians, the A/r^2 for solid angles has units of Steradians, Sr.

The total surface area of a sphere is $4\pi r^2$, so the total solid angle in all direction is 4π .

As we have defined the two cones to subtend the same solid angle at P , their areas scale as the square of their radii:

$$\frac{\Delta a_2}{\Delta a_1} = \frac{r_2^2}{r_1^2}$$

Hence:

$$\frac{\Delta q_2}{\Delta q_1} = \frac{\sigma \Delta a_2}{\sigma \Delta a_1} = \frac{r_2^2}{r_1^2}$$

The ratio of the two electric fields from each area is given by:

$$\frac{E_2}{E_1} = \frac{\Delta q_2 / r_2^2}{\Delta q_1 / r_1^2} = \frac{\Delta q_2}{\Delta q_1} \frac{r_1^2}{r_2^2} = 1$$

This means that the total E-field at P , and indeed at any point inside the hollow sphere, is zero.

3 Electric Flux

Electric flux is effectively the amount of E-field passing across a surface. It is defined in terms of a normal unit vector perpendicular to the surface, $\hat{n}$.

We get an expression for flux by multiplying the field strength by the area "seen" by the field. This is given by $A \cos \theta$ and for a constant E-field on a flat surface:

$$\text{flux} = \Phi_E = \underline{AE} \cdot \hat{n} = EA \cos \theta$$

More generally:

$$\Phi_E = \int \underline{E} \cdot \hat{n} dS = \int \underline{E} \cdot d\underline{S}$$

Where dS is an element of surface area.

4 Gauss' Law

Consider an imaginary closed surface, of any shape. Inside this surface contains a charge q .

The E-field at some element of this surface at distance r from the charge is given by:

$$\underline{E} = \frac{q}{4\pi\epsilon_0 r^2} \hat{r}$$

The flux out of this area element is:

$$d\Phi = \underline{E} \cdot d\underline{S} = \underline{E} \cdot \hat{n} dS = E \cos \theta dS$$

Hence:

$$d\Phi = \frac{q}{4\pi\epsilon_0 r^2} \cos \theta dS = \frac{q}{4\pi\epsilon_0} \frac{\cos \theta dS}{r^2} = \frac{q}{4\pi\epsilon_0} d\Omega$$

Where Ω is the solid angle.

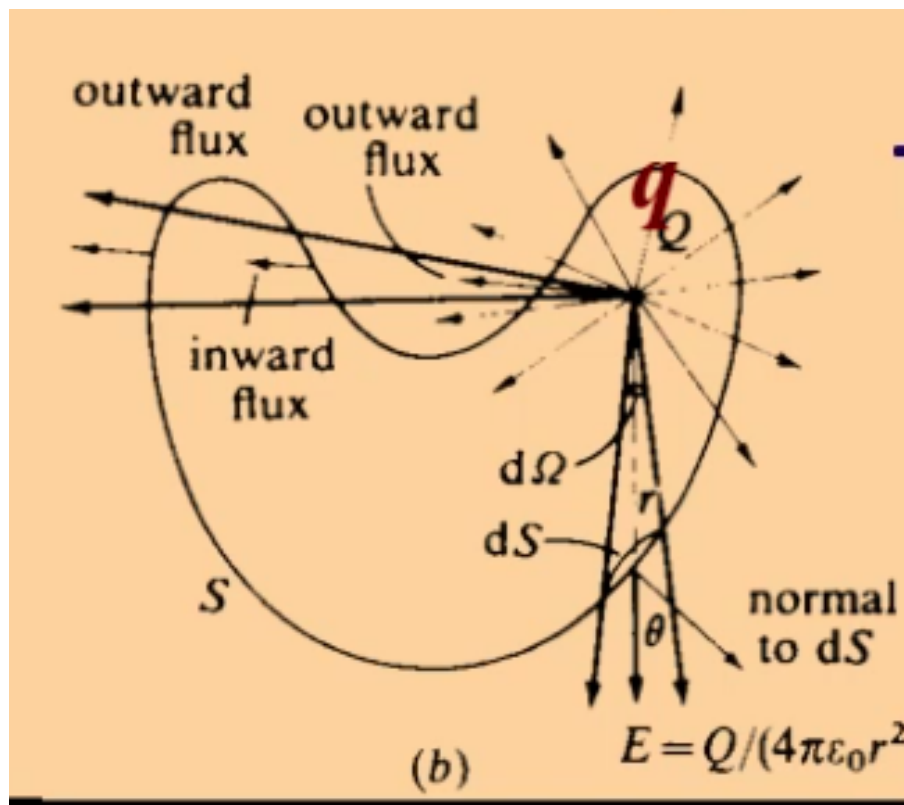


Figure 3.3

The total flux is therefore:

$$\Phi_E = \frac{q}{4\pi\epsilon_0} \int_S d\Omega = \frac{q}{4\pi\epsilon_0} 4\pi = \frac{q}{\epsilon_0}$$

Note that the proof/derivation is not needed, as solid angles are off-spec. Only the final result is needed.

This gives us Gauss' Law:

The net electric flux of $\underline{E}$ out of *any* closed surface, enclosing a total charge of Q_{enclosed} is:

$$\Phi_E = \int_S \underline{E} \cdot d\underline{S} = \frac{Q_{\text{enclosed}}}{\epsilon_0}$$

Where $Q_{\text{enclosed}} = \sum_i q_i$ for discrete charges or $Q_{\text{enclosed}} = \int_V \rho dV$ for distributed charge.

This becomes incredibly useful for problems where there is symmetry.

4.1 Example: Solid Sphere of Uniform Charge Density

Consider a sphere of radius R with charge Q distributed uniformly. We want to find the E-field inside the sphere, and the E-field outside the sphere.

We will set up an imaginary spherical Gauss surface of radius r and use symmetry:

$$\int_S \underline{E} \cdot d\underline{S} = \frac{Q_{\text{enclosed}}}{\epsilon_0}$$

Outside the Sphere: $r > R$

The E-field from a sphere and the normal vector of the surface are both radial. Hence, they are parallel. This means that $\theta = 0$ and:

$$\underline{E} \cdot d\underline{S} = E dS$$

Since the sphere is symmetrical, the E-field at a constant r must be constant anywhere around the sphere. Hence, E is constant and can be pulled out of the integral:

$$E \int_S dS = \frac{q_{\text{enclosed}}}{\epsilon_0}$$

$$E(4\pi r^2) = \frac{q_{\text{enclosed}}}{\epsilon_0}$$

$$E = \frac{Q}{4\pi\epsilon_0 r^2}$$

Which is Coulomb's law! Lets now consider the E-field inside the sphere.

Inside the Sphere: $r < R$

$$\int_S \underline{E} \cdot d\underline{S} = \frac{q_{\text{enclosed}}}{\epsilon_0}$$

As before, the LHS is $E4\pi r^2$. The RHS is now different, as a smaller volume of charge is enclosed by the Gauss surface. This new charge is given by the total charge times the ratio of the Gauss surface volume to the total volume:

$$\frac{q_{\text{enclosed}}}{\epsilon_0} = \frac{1}{\epsilon_0} \left(\frac{Qr^3}{R^3} \right)$$

Hence:

$$E4\pi r^2 = \frac{Qr^3}{\epsilon_0 R^3}$$

$$\implies E = \frac{Qr}{4\pi\epsilon_0 R^3}$$

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Lecture 4 - Electric Fields IV: Gauss' Law Examples

1 Solid Sphere with Non-Uniform Charge

What if we have a solid sphere with distributed charge where the charge distribution is a function of radius?

$$\rho(r) = \rho_0 r$$

Where r_0 is a constant (note: does not have dimensions of charge density).

We now have:

$$Q_{\text{enclosed}} = \int_0^r \rho(r) dV$$

Hence:

$$\int_S \underline{E} \cdot d\underline{S} = \frac{1}{\epsilon_0} \int_0^r \rho(r) dV$$

We divide the sphere into thin shells of infinitesimal thickness δr . The volume of each shell is:

$$\delta V = 4\pi r^2 \delta r$$

Hence:

$$\begin{aligned} Q_{\text{enclosed}} &= \int_0^r \rho(r) dV = \int_0^r \rho_0 r 4\pi r^2 dr \\ &= 4\pi \rho_0 \int_0^r r^3 dr = \pi \rho_0 r^4 \end{aligned}$$

Using Gauss' law:

$$\int_S \underline{E} \cdot d\underline{S} = \frac{\pi \rho_0 r^4}{\epsilon_0}$$

Considering the LHS and using symmetry:

$$\int_S \underline{E} \cdot d\underline{S} = \int_S E dS = E \int_S dS = E 4\pi r^2$$

Hence:

$$\begin{aligned} 4\pi E r^2 &= \frac{\pi \rho_0 r^4}{\epsilon_0} \\ \implies E &= \frac{\rho_0 r^2}{4\epsilon_0} \end{aligned}$$

2 E-Field Inside Uniform Charged Shell

We've already done this with solid angles, but it becomes much easier with Gauss' law. Since the Gauss surface sits inside the shell, none of the charged distributed across the shell is inside the surface. Therefore, $Q_{\text{enclosed}} = 0 \implies E = 0$

3 E-Fields in Conductors

A simple model of a conductor is where each atom has a free valance electron which is free to move throughout the conductor, in vast quantities.

If you place a conductive box inside a conductor, the electrons want to travel against the field. This leaves a positively charged nuclei, which drift in the opposite direction.

This causes the outside of the metal box to pick up a small charge, creating its own electric field which opposes the initial field. This continues until it perfectly cancels out the initial field, leaving no resultant field inside a conductor.

This is called a “Faraday Cage” and can be used to shield the interior of the conducting box from the E-field, i.e. to protect sensitive electronics.

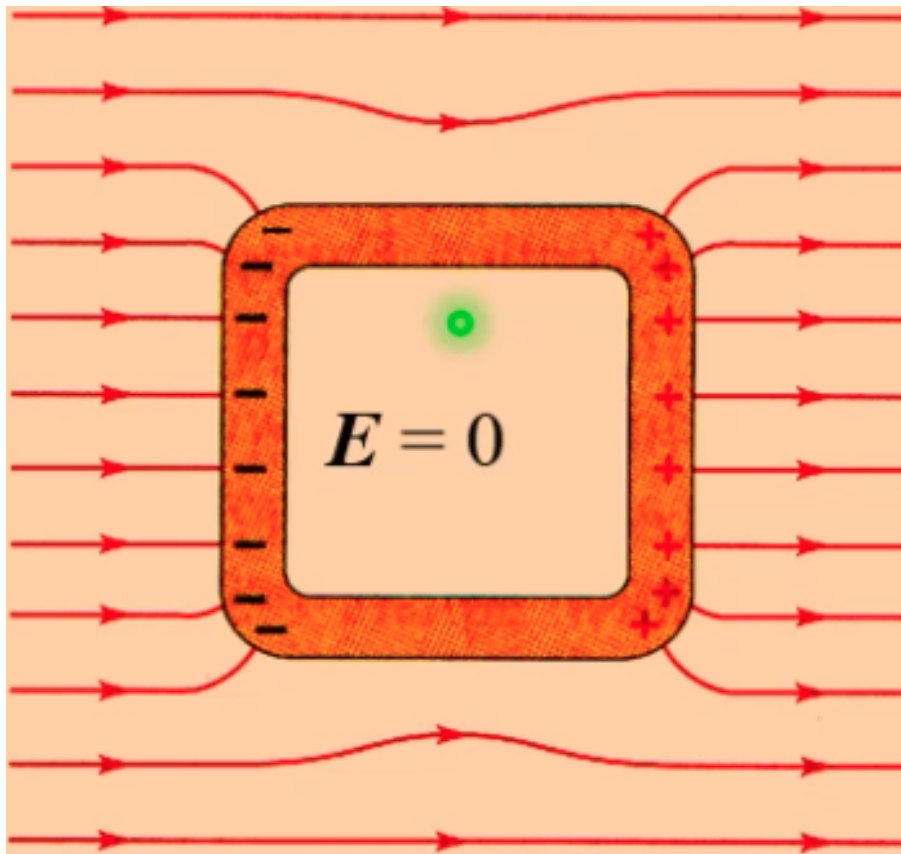


Figure 4.1

4 Infinite Charged Sheet

4.1 Non-Conducting Sheet

Consider an infinitely large plane of surface charge density σ . As the sheet is infinitely large, there's no edge effects and the E-field exists exclusively $\perp$ to the surface of the sheet.

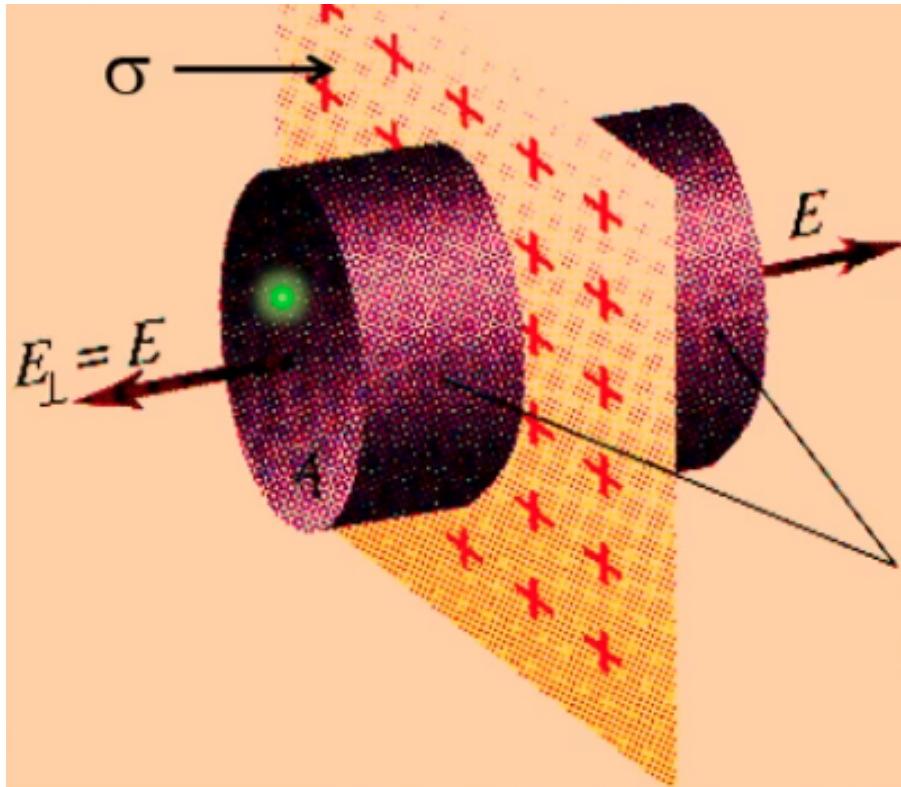


Figure 4.2

We can therefore take a cylindrical Gauss surface, and the only E-field will exist normally to the end circular faces (of cross-sectional area A). Hence, $\underline{E}$ and $d\underline{S}$ are parallel, and:

$$\int_S \underline{E} \cdot d\underline{S} = E \int_S dS$$

So:

$$\begin{aligned} E \int_S dS &= \frac{Q_{\text{enclosed}}}{\epsilon_0} \\ E(2A) &= \frac{\sigma A}{\epsilon_0} \\ \Rightarrow E &= \frac{\sigma}{2\epsilon_0} \end{aligned}$$

As previously derived! Again, this works for a finite plane where the plane is much larger (relatively) than the distance from the plane, so edge effects can be disregarded.

4.2 Conducting Sheet

As the sheet is now conducting, there is no E-field inside the conductor, and charge is distributed to build up a charge density (σ) on both surfaces of the sheet.

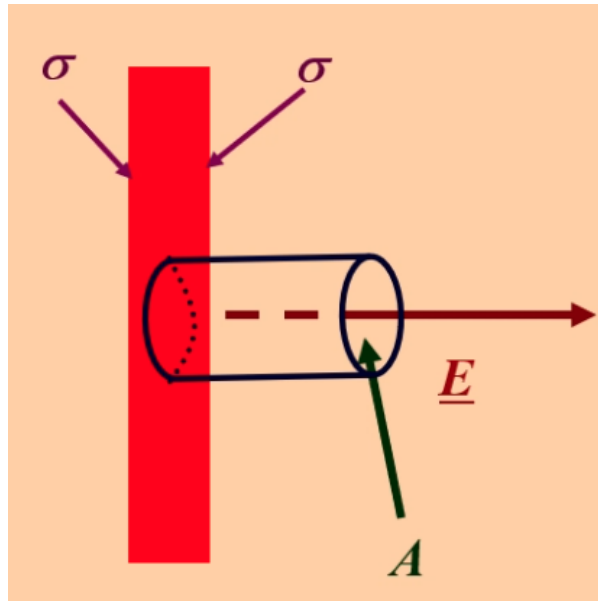


Figure 4.3

Applying Gauss' law gives:

$$EA = \frac{\sigma A}{\epsilon_0}$$

$$\Rightarrow E = \frac{\sigma}{\epsilon_0}$$

Which is a factor of two different from the non-conducting case. Note that this doesn't arise because we only took a Gauss surface on one side. If we took a Gauss surface through the sheet (in the same manner as the non-conducting case), we would have two sets of charge distributed, one on each side, both with charge density σ . We would therefore have:

$$E(2A) = \frac{2\sigma A}{\epsilon_0}$$

Which yields the same result.

5 E-Field Between Charged Conducting Plates

Consider a capacitor-like setup with two parallel charged metal plates.

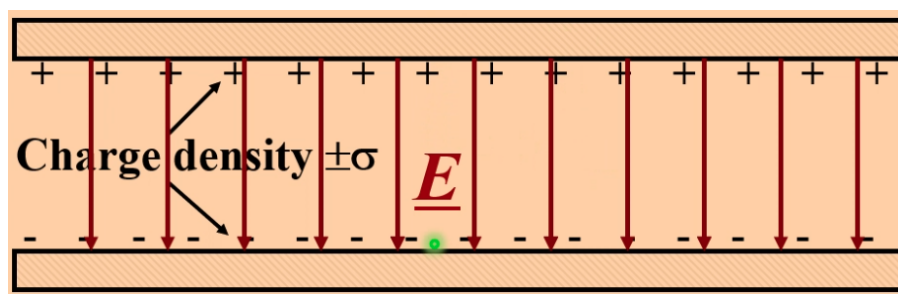


Figure 4.4

While a full derivation will be covered later, the E-field given by this is the same as for an infinite sheet:

$$E = \frac{\sigma}{\epsilon_0}$$

6 Infinite Line of Charge

Consider an infinitely long line charge, of charge density λ . As its infinitely long, there's no edge effects and the electric field must be radial.

We can place a cylindrical Gaussian surface where the axis is parallel to the line, of length L and radius r .

Since the field is radial, it acts in the same direction as the normal vector to the cylinder surface. By symmetry, the E-field is constant for fixed radius, so:

$$\int_S \underline{E} \cdot d\underline{S} = E \int_S dS = E(2\pi rL)$$

$$\frac{Q_{\text{enclosed}}}{\epsilon_0} = \frac{\lambda L}{\epsilon_0}$$

Hence:

$$E(2\pi rL) = \frac{\lambda L}{\epsilon_0}$$

$$E = \frac{\lambda}{2\pi\epsilon_0 r}$$

7 No Contained Charge

What if the Gaussian surface sits within an E-field, but the charged creating the field is not enclosed in the surface?

This could be thought of as implying by Gauss' law that:

$$\oint \underline{E} \cdot d\underline{S} = 0$$

$$\implies E = 0?$$

But this is not the case. Here, the symmetry that we've relied upon isn't here, as $\underline{E} \nparallel d\underline{S}$ and is not constant around the surface. Hence, we can't nicely simplify the problem and the first integral becomes unsolvable (by us).

If the charge falls outside of the Gauss surface, the net flux through the surface is zero.

Thu 02 Feb 2026 11:00

Lecture 5 - Electric Fields IV: Gauss Examples and Electric Potential

1 Non-Conducting Infinite Charged Parallel Plates

Consider two non-conducting infinite planes of charge with charge density σ .

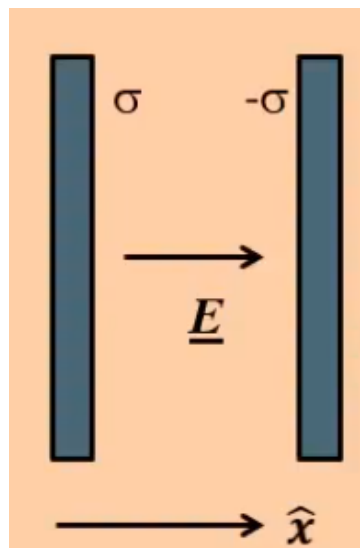


Figure 5.1

We have derived that the surface charge from the LHS plate gives field:

$$\underline{E}_L = \frac{\sigma}{2\epsilon_0} \hat{x}$$

The RHS plate gives:

$$\underline{E}_R = \frac{-\sigma}{2\epsilon_0} \hat{n} = \frac{\sigma}{2\epsilon_0} \hat{x}$$

Where $\hat{n}$ is the unit vector normal to the RHS surface. By superposition, the total field between the plates is:

$$\underline{E} = \underline{E}_L + \underline{E}_R = \frac{\sigma}{\epsilon_0} \hat{x}$$

This is exactly the same as for two conducting plates, per last lecture.

1.1 Back to Conducting Plates

Consider two oppositely charged conducting plates a long way apart. They have charges on each surface.

As the two plates move closer together, their E-fields attract the charges on each other. This causes movement of the charges, until all the charges move towards the front faces. This causes the effective surface charge density to double, hence:

$$E = \frac{\sigma_i}{\epsilon_0} + \frac{\sigma_i}{\epsilon_0} = \frac{2\sigma_i}{\epsilon_0} = \frac{\sigma}{\epsilon_0}$$

Where σ_i is the initial charge densities on the surface of each plate, and σ is the final charge density effective on the inner surfaces. There is therefore no charges (and hence $E = 0$) on the outer surfaces. Note that this is only true when the plates have equal and opposite charge densities.

2 Work Done by an E-Fields

Consider a particle moving in an E-field. Across some infinitesimal distance $\delta \underline{l}$, over which the particle experiences force $\underline{F}$, the infinitesimal element of work done is:

$$\delta W = \underline{F} \cdot \delta \underline{l}$$

As the E-field by definition carries units of force per unit charge, the force is given by:

$$\delta W = q \underline{E} \cdot \delta \underline{L}$$

The work done by an E-field represents a decrease in electric potential energy, $\delta U = -\delta W$:

$$\delta U = -q \underline{E} \cdot d\underline{L}$$

If the charge is positive, δU is negative and energy decreases. If the charge is negative, δU is positive and energy increases.

If the E-field varies wrt space, and the particle moves from point A to point B, we need to integrate to find the total change:

$$U_B - U_A = -q \int_a^b \underline{E} \cdot d\underline{l}$$

We want to be able to define this on a per unit charge basis.

V is the potential energy of the system *per unit charge*.

$$V = \frac{U}{q}$$

$$V_b - V_A = - \int_a^b \underline{E} \cdot d\underline{l}$$

It is a property of a point everywhere in an E-field. **It is a scalar**. The units are Joules per Coulomb, i.e. Volts.

The electric potential at distance r from a point charge q is:

$$\underline{E} = \frac{q}{4\pi\epsilon_0 r^2} \underline{r}$$

$$V(\infty) - V(r) = - \int_{\infty}^r \underline{E} \cdot d\underline{r} = \frac{q}{4\pi\epsilon_0 r}$$

This is the Coulomb potential, which can be positive or negative (as q can be positive or negative).

As it's potential per unit charge, the U of a charge q_0 at distance r from charge q is:

$$U(r) = V(r)q = \frac{1}{4\pi\epsilon_0} \frac{qq_0}{r}$$

And:

$$F = - \frac{dU(r)}{dr} = \frac{1}{4\pi\epsilon_0} \frac{qq_0}{r^2}$$

3 Multiple Charges

What if we have multiple charges? We simply sum across them:

$$V = \frac{1}{4\pi\epsilon_0} \sum_i \frac{q_i}{r_i}$$

Where r_i is the distance from the i th charge q_i to the point at which we're evaluating the potential.

4 Electric Potential Difference

The electric potential difference (AKA voltage) between two points is:

$$V_B - V_A = \frac{U_B - U_A}{q} = - \int_a^b \underline{E} \cdot d\underline{l}$$

We can use this to get that Newtons per Coulomb and Volts per Metre are dimensionally equivalent units, and we commonly use the latter for describing electric fields.

5 E-Field Lines

The lines of an E-field trace the path that a positive test charge would follow under the action of the electrostatic force. If the positive test charge were released, it would accelerate along the field lines, from high potential to low potential.

We commonly overlay perpendicular (to $\underline{E}$) lines on a field diagram, representing surfaces where V is a constant. These are called equipotentials and are always perpendicular to $\underline{E}$.

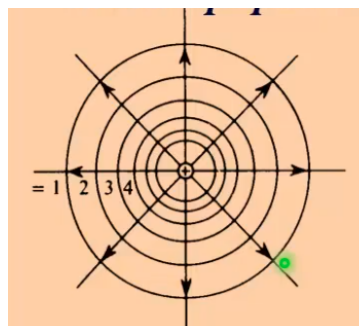


Figure 5.2

6 E from V

Because V is a scalar quantity, its often easier to calculate than E . There are some cases where its therefore easiest to find V and use this to determine E . We start from:

$$V = - \int \underline{E} \cdot d\underline{l}$$

And differentiate both sides:

$$\underline{E} = - \left(\frac{\partial V}{\partial x} \hat{i} + \frac{\partial V}{\partial y} \hat{j} + \frac{\partial V}{\partial z} \hat{k} \right) = - \underline{\nabla} V$$

Or in polar coordinates:

$$\underline{E} = - \left(\frac{\partial V}{\partial r} \hat{r} + \frac{1}{r} \frac{\partial V}{\partial \theta} \hat{\theta} \right)$$

Where the $1/r$ term arises to ensure dimensional correctness. Often, we have nice examples where we're only dealing with the radial component, so have:

$$E(r) = -\frac{dV}{dr}\hat{r} = \frac{q}{4\pi\epsilon_0 r^2}\hat{r}$$

7 Conservatism

The $\underline{E}$ -field is a conservative field. This means that if the charge returns to its original position, regardless of the route taken, no net work is done:

$$\oint \underline{E} \cdot d\underline{l} = 0$$

This equivalently means that the change in potential between two points is the same whichever path between the two points is taken.

Thu 26 Feb 2026 11:00

Lecture 12 - Currents and Magnetic Force

Today we start Part II, Magnetism:

- Definition of current and current density.
- Magnetic force on a moving charge.
- The Lorentz Force.
- Field lines for a magnetic field.

1 Current

1.1 Definition of Current

Suppose a current carries a current I , this is defined in terms of the rate of flow of charge past a given cross section:

$$I = \frac{dQ}{dt}$$

The SI unit of current is the *ampere*, defined as the flow of one coulomb per second.

1.2 Nature of Current

There's two experimentally observed effects of a current flow:

- Heating
- Creation of magnetic fields.

Consider a current through a conductor. This is caused by electrons moving (with average drift velocity v) as a result of the induced E-field.

In some time, they travel $v\Delta t$ through the cross section, and if this cross section has area A they sweep out an area of $Av\Delta t$.

If a unit volume has n conducting free electrons, the number of charges is given by:

$$N = n(Av\Delta t)$$

And hence the total charge is:

$$\Delta Q = nAv\Delta t(-e)$$

Hence:

$$\hat{i} = \frac{\Delta Q}{\Delta t} = -nAev$$

1.3 Current Density

More usefully, we talk about the current per unit cross-section, which is the current density J :

$$\hat{j} = \frac{\hat{i}}{A} = -nev$$

The negative sign arises as we define current as flowing from positive to negative. Electrons, being negatively charged, actually physically flow from negative to positive. Current flow and actual electron flow are therefore in the opposite direction. This is because current was initially understood and defined before the electron was defined.

Typically, $v < 1\text{mms}^{-1}$ when a current is flowing.

2 Magnetism

2.1 Magnetic Force on a Charge

A magnetic field exerts a force on a moving charge that is present in the magnetic field.

Suppose a particle of charge $+q$ moving with some velocity $\underline{v}$ in a magnetic field $\underline{B}$ experiences some force $\underline{F}_m$.

We cannot derive from first principles, but experimentally have observed:

- $\underline{F}_m \perp \underline{v}, \underline{B}$.
- $\underline{F}_m \propto \underline{v}$.
- $\underline{F}_m \propto \underline{B}$.
- $\underline{F}_m \propto q$

If θ is the angle between the velocity and the B-field direction, we have:

$$\underline{F}_m = Bqv \sin \theta$$

In vector form:

$$\underline{F}_m = q\underline{v} \wedge \underline{B}$$

Here, $\underline{B}$ has the unit Tesla, T. In base units this is $\text{NC}^{-1}\text{m}^{-1}\text{s}$. A one Tesla magnetic field is really quite powerful, so we also define the Gauss, $1\text{G} = 10^{-4}\text{T}$ to move to a nicer scale.

By definition, if a 1C charge moving at 1m/s perpendicular to a magnetic field experiences a force 1N, the field is 1T.

- Earth's magnetic field is $\sim 0.5\text{G}$
- Poles of a large electromagnetic: $\sim 2\text{T}$.
- Surface of a neutron star: $\sim 10^8\text{T}$.
- The maximum pulsed magnetic field that can create in a lab: $\sim 50\text{T}$.

The Earth's field is believed to be generated by electron currents in the iron alloys in its core. It's not completely understood yet, but convection currents causing these conductive alloys to flow and move is the working theory. This movement causes the North and South poles to swap places on average every 300,000 years.

2.2 The Lorentz Force

In a region where we have both an E-field and a B-field, we have the total force as the vector sum of both:

$$\underline{F} = q(\underline{E} + \underline{v} \wedge \underline{B})$$

This is called the Lorentz force.

Suppose Earth's magnetic field is given by:

$$\underline{B} = B \cos 70^\circ \hat{j} - B \sin 70^\circ \hat{k} \quad \text{where: } B = 5 \times 10^{-5}\text{T}$$

A proton moves in this field with:

$$\underline{v} = 10^7 \hat{j} \text{ms}^{-1}$$

$$\begin{aligned}
 \underline{F}_m &= +e\underline{v} \wedge \underline{B} = 1.6 \times 10^{19} \times 10^7 \times \det \begin{bmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 1 & 0 \\ 0 & B_y & B_z \end{bmatrix} \\
 &= 1.6 \times 10^{-12} [\hat{i}(B_z - 0) + \hat{j}(0 - 0) + \hat{k}(0 - 0)] \\
 &= 1.6 \times 10^{-12} B_z \hat{i} \\
 &= -1.6 \times 10^{-12} \times 5 \times 10^{-5} \sin 70^\circ \hat{i} \\
 &= -7.5 \times 10^{-17} \hat{i} \text{N}
 \end{aligned}$$

2.3 Magnetic Field Lines

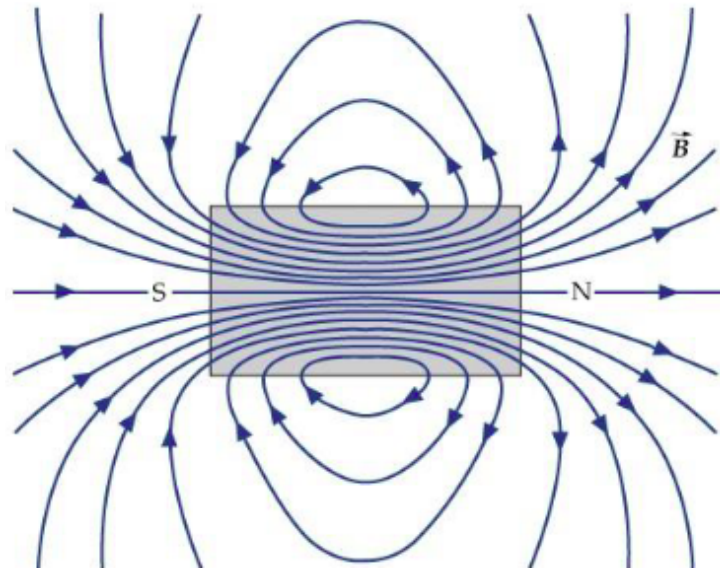


Figure 6.1: Magnetic Field Lines Around a Bar Magnetic

Magnetic field lines are different to electrical field lines. They do not point in the direction of force and travel out of North poles and into South poles. Magnetic field lines are always continuous so there is no magnetic monopoles.

Physicists have searched for a potential magnetic monopoles, and the current upper limit for the number of magnetic monopoles per body is $< 10^{-29}$.

The tangent to a field line at some point gives the direction of B at that point. The number of field lines drawn per unit cross section $\propto B$.

2.4 Magnetic Flux

The magnetic flux $\Delta\phi_B$ has the same definition as electric flux in an E-field. For a small area ΔA :

$$\Delta\phi_B = B \cos \theta \Delta A$$

Where θ is the angle between the B-field and the surface normal vector.

The total magnetic flux is therefore the integral of this:

$$\phi_B = \int_S \underline{B} \cdot d\underline{S}$$

As there is no monopoles, and exiting flux must also return. Therefore, the net flux over any enclosed surface is zero:

$$\int_S \underline{B} \cdot d\underline{S} = 0$$

While this does form one of Maxwell's equations and is very useful for EMII next year, it's not useful in EMI...

Thu 05 Mar 2026 11:00

Lecture 14 - Magnetic Fields from Currents

Recap of last lecture:

- Special cases of the magnetic force.
- Force on a current carrying conductor:

$$\underline{F} = I \underline{L} \wedge \underline{B}$$

- Current loops and magnetic dipoles:

$$\underline{\mu} = I \underline{A}$$

- Torque on a magnetic dipole in a B-field

$$\underline{\tau} = \underline{\mu} \wedge \underline{B}$$

- Potential energy of a magnetic dipole in a B-field:

$$U = -\underline{\mu} \cdot \underline{B}$$

1 Magnetic Fields from a Moving Charge (Current)

Danish physicist Hans Christian Oersted in 1820 demonstrated that a magnetic field exists around a current-carrying wire.

This was the first connection between electric and magnetic phenomena.

Consider a B-field at point $\underline{r}$ from a charge q , moving with velocity $\underline{v}$. The angle between $\underline{r}$ and $\underline{v}$ is given by ϕ .

The induced B-field is perpendicular to both $\underline{u}$ and $\underline{r}$ and is given by:

$$B \propto \frac{qv \sin \phi}{r^2}$$

In vector form, we have:

$$\underline{B} \propto \frac{q}{r^2} \underline{u} \wedge \hat{r}$$

In SI units, the constant of proportionality (involving “permeability of free space”, μ_0) is such:

$$\underline{B} = \frac{\mu_0}{4\pi} \frac{q}{r^2} \underline{u} \wedge \hat{r} \quad \mu_0 = 4\pi \times 10^{-7} \text{TmA}^{-1}$$

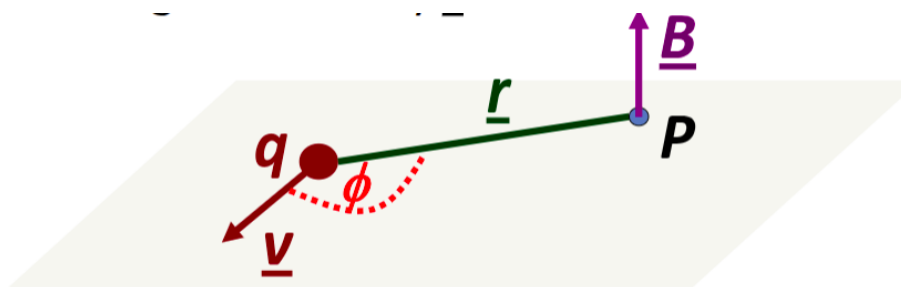


Figure 7.1

In a more complex 3D case, this looks as such:

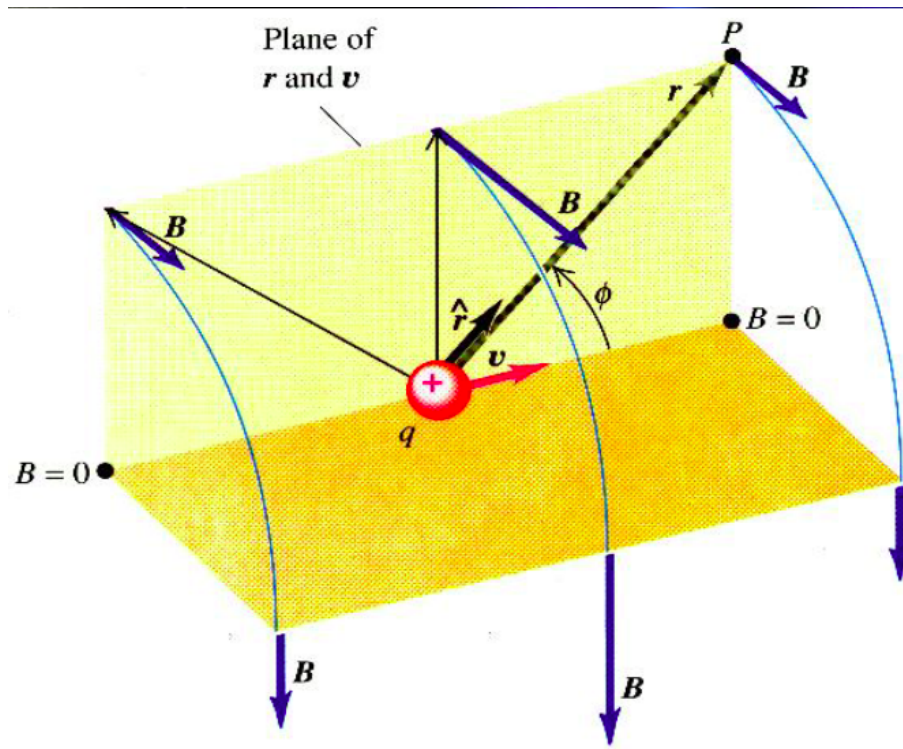


Figure 7.2

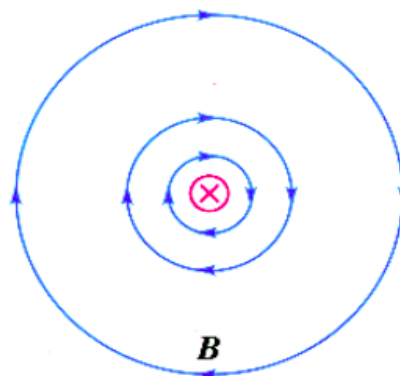


Figure 7.3

The B-field sweeps out a screw shape around the motion of direction. The direction is given by the right hand rule, where a “thumbs up” with the thumb pointing in the direction of the motion of the charge indicates the direction with the curl of the fingers.

1.1 Currents

Consider a wire with a current flow through it. The total charge flowing through some infinitesimal element in a unit time is:

$$I = \frac{qv}{\delta L}$$

Hence:

$$qv = \frac{qv}{\delta L} \delta L = I \delta L \quad \text{where } \delta L \text{ is in the direction of current.}$$

Plugging this in, we get the Biot-Savat Law:

$$\delta \underline{B} = \frac{\mu_0}{4\pi} \frac{I \delta \underline{L} \wedge \hat{r}}{r^2}$$

We can split our object into infinitely many infinitesimal elements and integrate across them all to find the total B-field. We want to:

- Define $d\mathbf{B}$ in terms of a single variable.
- Integrate between the relevant limits.
- Be careful about the directions of the included vector quantities.
- Use symmetry where possible.

2 Current Loops

Consider a current loop with a clockwise current flow. The centre of the loop's circle is P , with a radius from P of R .

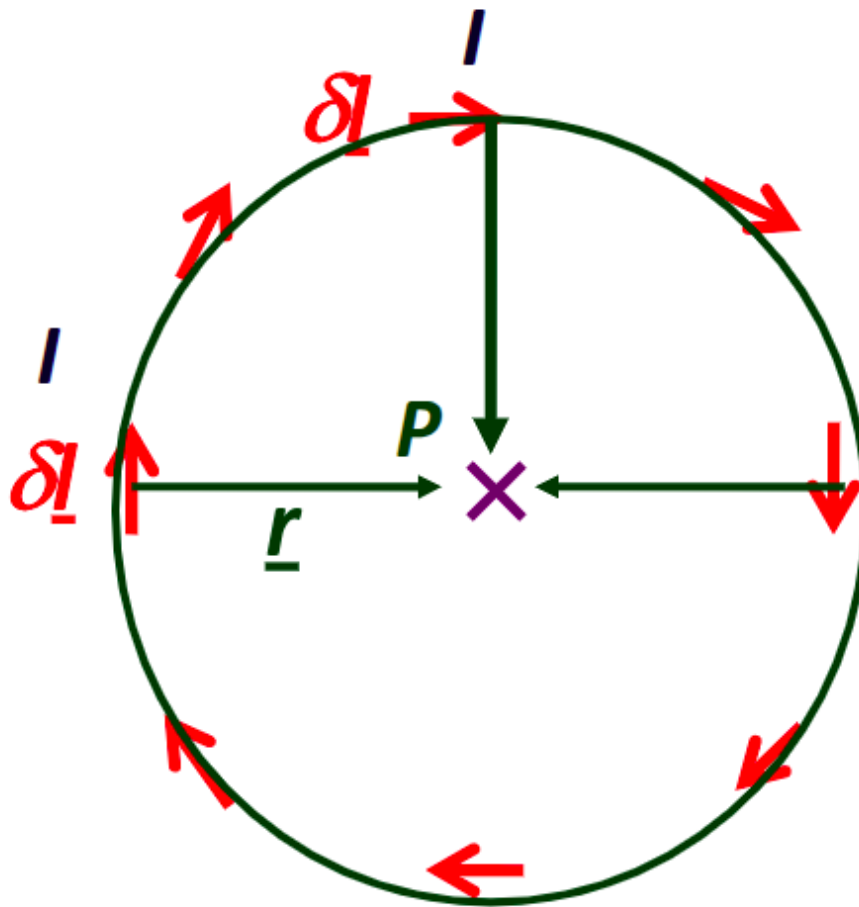


Figure 7.4

The Biot-Savat law gives:

$$\begin{aligned}\delta\mathbf{B} &= \frac{\mu_0}{4\pi} I \frac{\delta\mathbf{l} \wedge \hat{\mathbf{r}}}{r^2} \\ \delta B &= \frac{\mu_0}{4\pi} I \frac{\delta L}{r^2} \\ B &= \frac{\mu_0}{4\pi} \frac{I}{r^2} \oint dL = \frac{\mu_0}{4\pi} \frac{I}{4\pi} \times 2\pi r \\ B &= \frac{\mu_0 I}{2r}\end{aligned}$$

3 Finite Wire

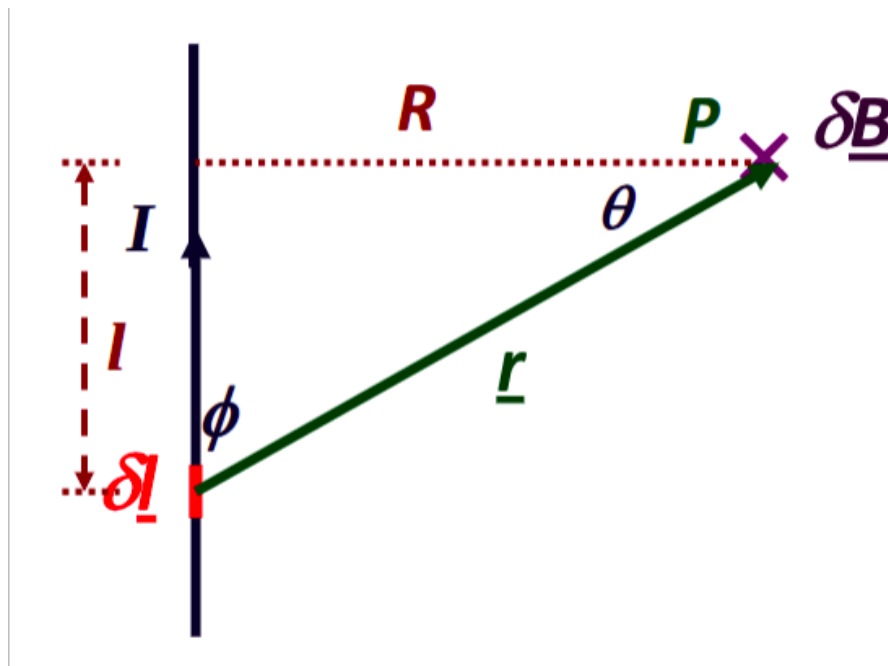


Figure 7.5

We want to find the B-field from the line of current. The B-field is going into the page at P, given by the cross. The Biot Savat Law again gives:

$$\underline{\delta B} = \frac{\mu_0 I}{4\pi r^2} \underline{\delta L} \wedge \hat{r}$$

As $\hat{r}$ is a unit vector, the magnitude is given with the magnitude of L and the sine of phi:

$$\delta B = \frac{\mu_0 I}{4\pi r^2} \delta L \sin \phi \quad (1)$$

We want this in terms of θ instead, and given these two angles form a right angled triangle we have:

$$\sin \phi = \sin \left(\frac{\pi}{2} - \theta \right) = \cos \theta \quad (2)$$

Hence:

$$\delta B = \frac{\mu_0}{4\pi r^2} \delta L \cos \theta$$

By trig we simply have:

$$r \cos \theta = R \implies r = \frac{R}{\cos \theta} \quad (3)$$

$$\begin{aligned} \tan \theta = \frac{L}{R} &\implies L = R \tan \theta \implies \frac{dL}{d\theta} = \frac{R}{\cos^2 \theta} \\ &\implies \delta L = \frac{R}{\cos^2 \theta} \delta \theta \end{aligned} \quad (4)$$

Inserting (2), (3), (4) back into (1) gives:

$$\delta B = \frac{\mu_0 I}{4\pi} \times \frac{\cos^2 \theta}{R^2} \times \cos \theta \times \frac{R}{\cos^2 \theta} \delta \theta$$

Hence:

$$\delta B = \frac{\mu_0 I}{4\pi} \frac{1}{R} \cos \theta \delta \theta$$

$$B = \frac{\mu_0 I}{4\pi R} \int_{\theta_1}^{\theta_2} \cos \theta d\theta$$

Where θ_1, θ_2 depend the angles given by the length of the wire.

$$B = \frac{\mu_0 I}{4\pi R} (\sin \theta_2 - \sin \theta_1)$$

If the wire was infinitely long, $\theta_2 = \pi/2$ and $\theta_1 = -\pi/2$ (technically a limit as $L \rightarrow \infty$). This gives us:

$$B = \frac{\mu_0 I}{2\pi R}$$

4 More Complex Loops

Consider a current loop, with a set of axes centred at the middle of the loop. The loop sits in the yz plane, and we step out some perpendicular distance x in the x plane.

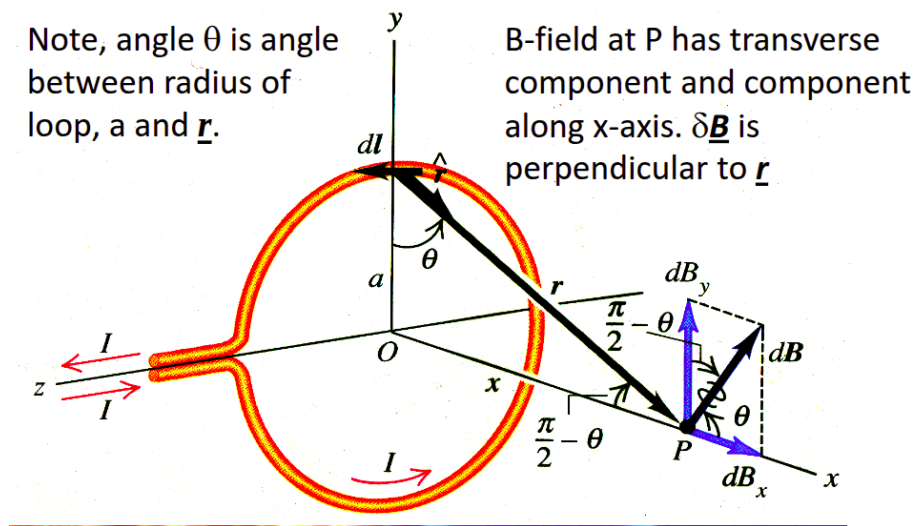


Figure 7.6

We step some distance δL along this loop, and the vector from the point we're investigating to B-field at from this δL is $\underline{r}$, at angle θ .

We can consider symmetry to see that transverse components to the axis will cancel each other out and therefore do not need considering:

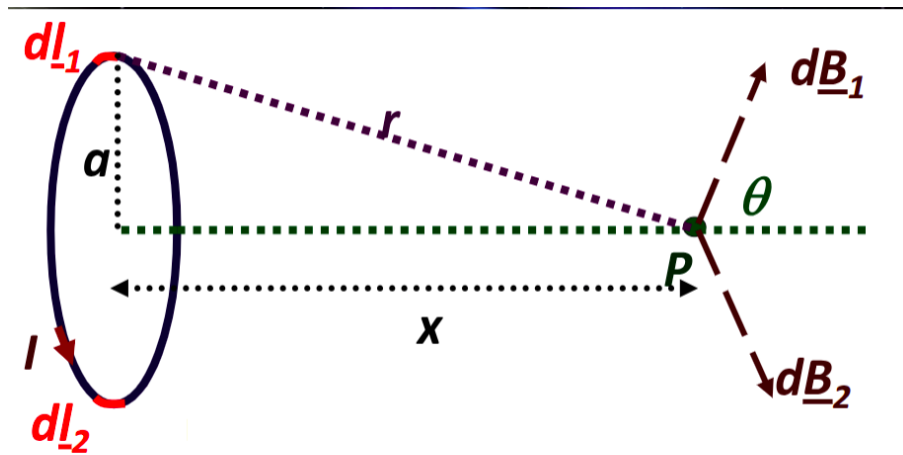


Figure 7.7

Therefore all of the components of the B-field in the yz planes will cancel out, and the only resultant B-field is in the x-direction. We will fully evaluate this on Monday.

Mon 09 Mar 2026 11:00

Lecture 15 - Ampere's Law

1 Biot-Savat: Complex Loops Continued

Continuing from the end of last lecture, we found by symmetry that the yz components cancel, leaving only an x component for any point P along the x -axis.

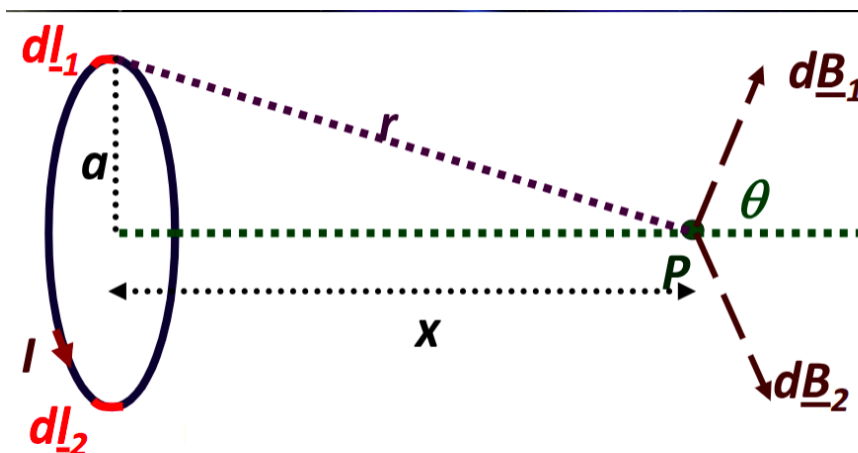


Figure 8.1

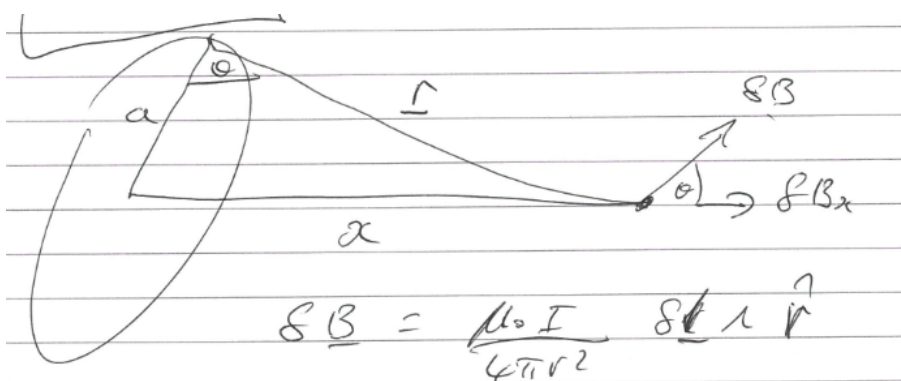


Figure 8.2

Taking magnitudes (and noting that $\hat{r}$ is a unit vector):

$$\delta B = \frac{\mu_0 I}{4\pi r^2} \delta l$$

Resolving into the x -direction, we get that:

$$\delta B_x = \frac{\mu_0 I}{4\pi r^2} \delta l \cos \theta \quad (1)$$

And as we have a right angled triangle:

$$r^2 = a^2 + x^2 \quad (2)$$

$$\cos \theta = \frac{a}{r} \implies \cos \theta = \frac{a}{(a^2 + x^2)^{(1/2)}} \quad (3)$$

Inserting 2 and 3 into 1:

$$\delta B_x = \frac{\mu_0 I}{4\pi} \times \frac{a}{(a^2 + x^2)^{(3/2)}} \delta l$$

$$B_x = \frac{\mu_0 I}{4\pi} \times \frac{a}{(a^2 + x^2)^{(3/2)}} \oint \delta l$$

$$B_x = \frac{\mu_0 I}{4\pi} \times \frac{a}{(a^2 + x^2)^{(3/2)}} 2\pi a$$

$$B_x = \frac{\mu_0 I}{2} \frac{a^2}{(a^2 + x^2)^{(3/2)}}$$

Taking limits, we can observe that at the centre of the loop (where $x = 0$):

$$B_0 = \frac{\mu_0 I}{2} \frac{a^2}{(a^2)^{(3/2)}} = \frac{\mu_0 I}{2a}$$

And when $x \gg a$, we have:

$$B_x = \frac{\mu_0 I a^2}{2x^3}$$

However, we can simplify using the magnetic dipole moment $\mu_m = I \times \text{area} = I\pi a^2$, hence:

$$B_x = \frac{\mu_0}{4\pi} \frac{2\mu_m}{x^3}$$

This is strikingly similar for the expression for an electric dipole:

$$E_x = \frac{1}{4\pi\epsilon_0} \frac{2p}{x^3}$$

2 E- and B-Fields

We have already determined the following, for E- and B-fields respectively:

$$\int_S \underline{E} \cdot d\underline{S} = \frac{Q}{\epsilon_0} \quad (E)$$

$$\int_S \underline{B} \cdot d\underline{S} = 0 \quad (B)$$

We also know that the E-field is conservative, hence no work is done if a charge returns to its original position:

$$\oint \underline{E} \cdot d\underline{l} = 0$$

Is there an equivalent of this for B-fields?

Consider a circular path of a B-field created around an infinite line of current at radial distance r . By symmetry, $\underline{B}$ and $d\underline{l}$ are parallel, and are constant for a fixed r . Hence:

$$\oint \underline{B} \cdot d\underline{l} = \oint B dl = B \oint dL = B 2\pi r$$

We have also determined that:

$$B = \frac{\mu_0 I}{2\pi r} \implies \oint \underline{B} \cdot d\underline{l} = B 2\pi r = \frac{\mu_0 I}{2\pi r} \times 2\pi r = \mu_0 I$$

This gives us Ampere's Law:

$$\oint \underline{B} \cdot d\underline{l} = \mu_0 I_{\text{enclosed}}$$

2.1 Example

Consider a long solid cylinder carrying a uniformly distributed current. The cylinder has radius R and length

Thu 12 Mar 2026 11:00

Lecture 16 - Electromagnetic Induction

1 Ampere's Law Continued

2 Magnetic Inductance

Faraday suggested that, if currents induce magnetic fields, that the opposite might be possible. Could a magnetic field be used to induce a current? He found that whenever field lines move or change, an electric field is induced, which charges in the field respond to.

3 Motion of a Conductor in Magnetic Field

4 Induced E-Fields

5 Faraday's Law